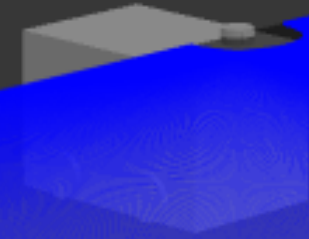


Autonomous Robot Assignment DEMO



James McMillan

202202995

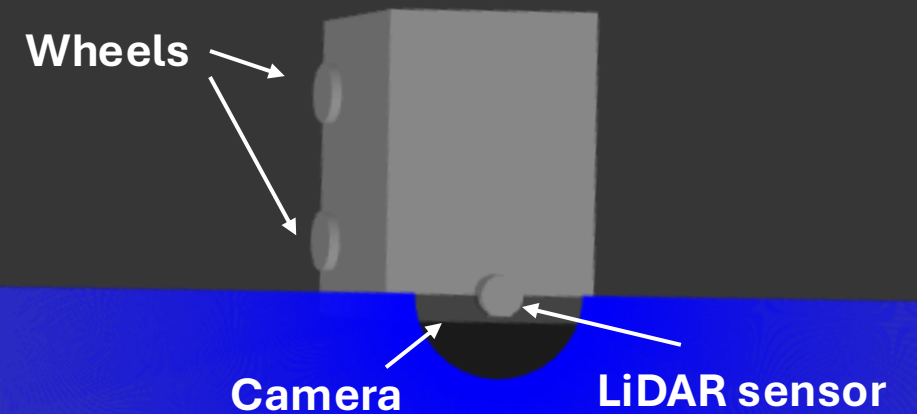
LIMO Robot



LIMO Robot

Key components:

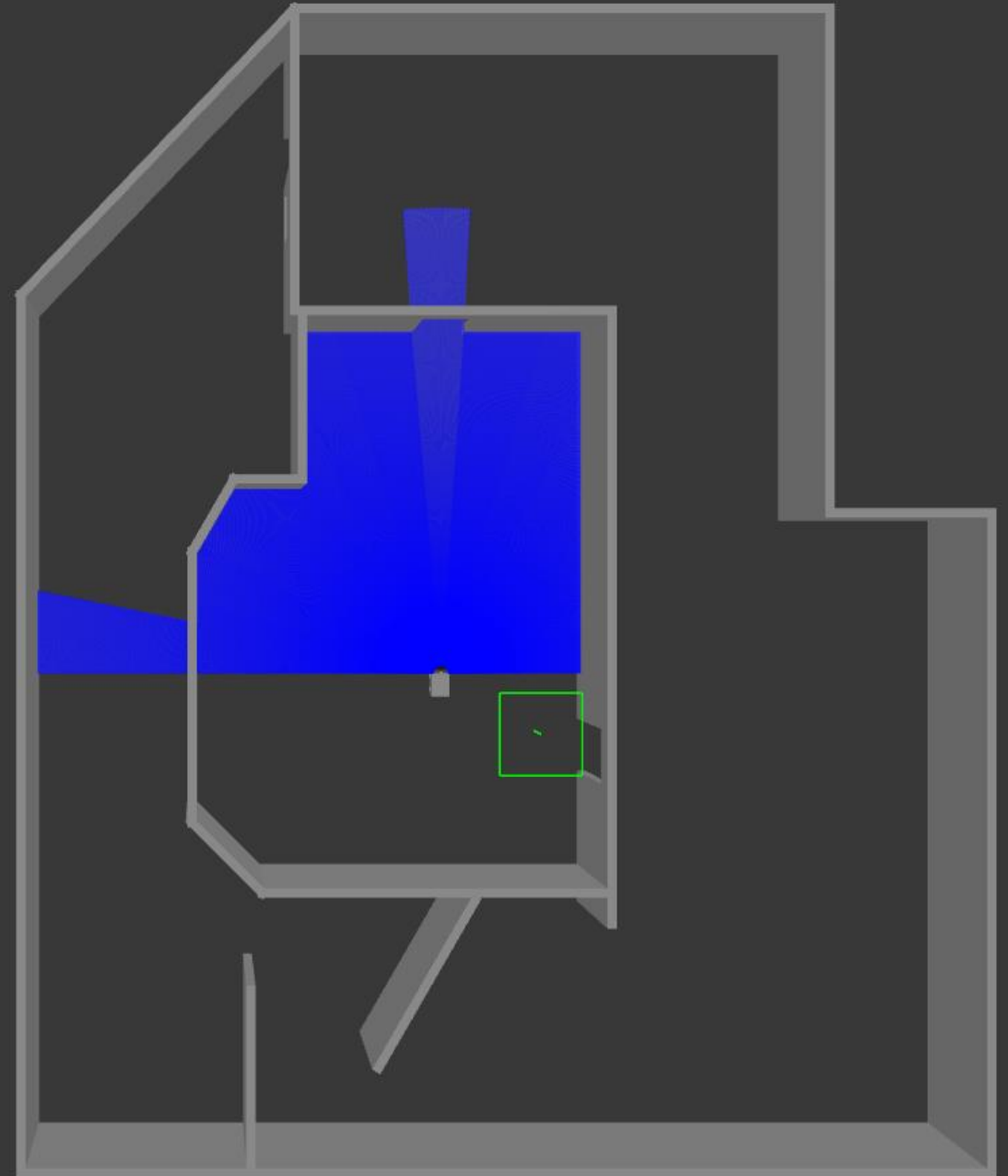
- Wheels (x4)
- Lidar sensor
- Camera
- Main body



Within the simulation

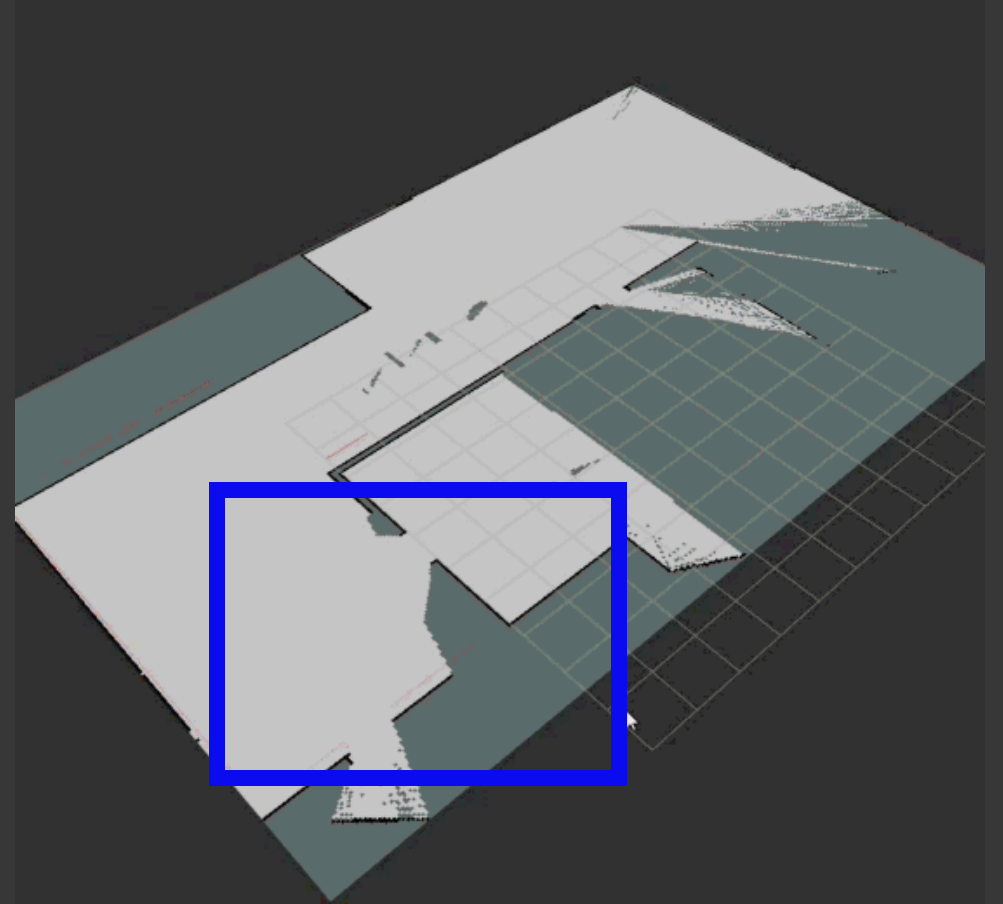
Simulated building

- Simulated building with a size of approximately 19 meters by 16.5 meters.
- Self-contained – no escape for the robot.
- Features:
 - Tight corners
 - Doorways
 - Narrow passages



SLAM

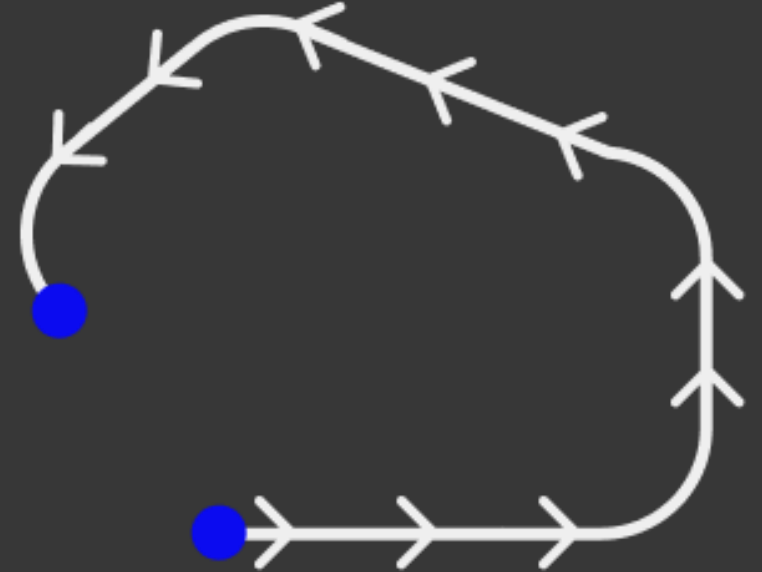
- Simultaneous location and mapping
- Process to generate a map of the robots' surroundings.
- Uses the data from the LiDAR sensor. Extracting features of the surroundings.
- Uses a POSE graph to work out where the robot is.
- The robot explores the building being controlled by tello, absorbing the sensor data.
- Camera feed used for manual validation.



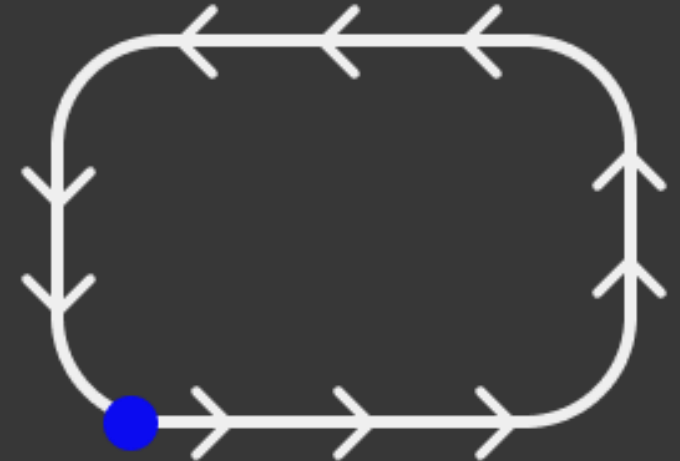
Loop Closure

- Activated once the robot returns to a position it has already been to.
- Corrects drift within the map
- Reduces/eliminates warping within the generated map.

Before



After



Navigation

Dynamic window approach

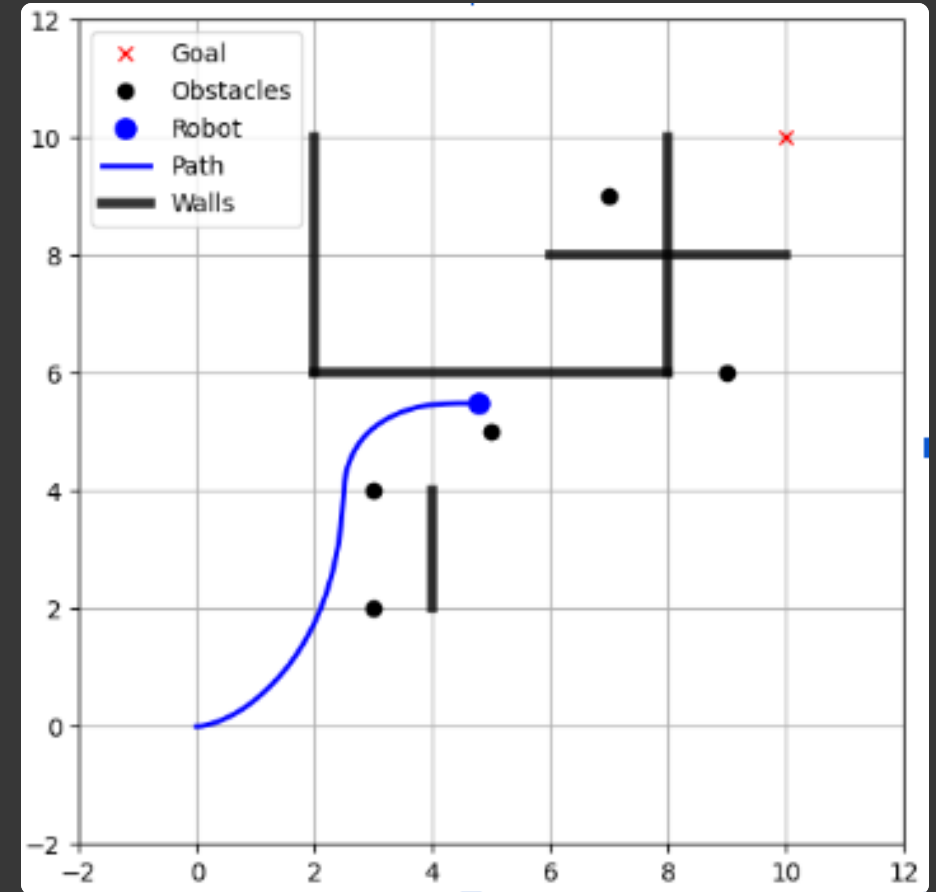
Selects safe, feasible velocity commands by evaluating how the robot can move within its dynamic limits.

A Star Path planning algorithm

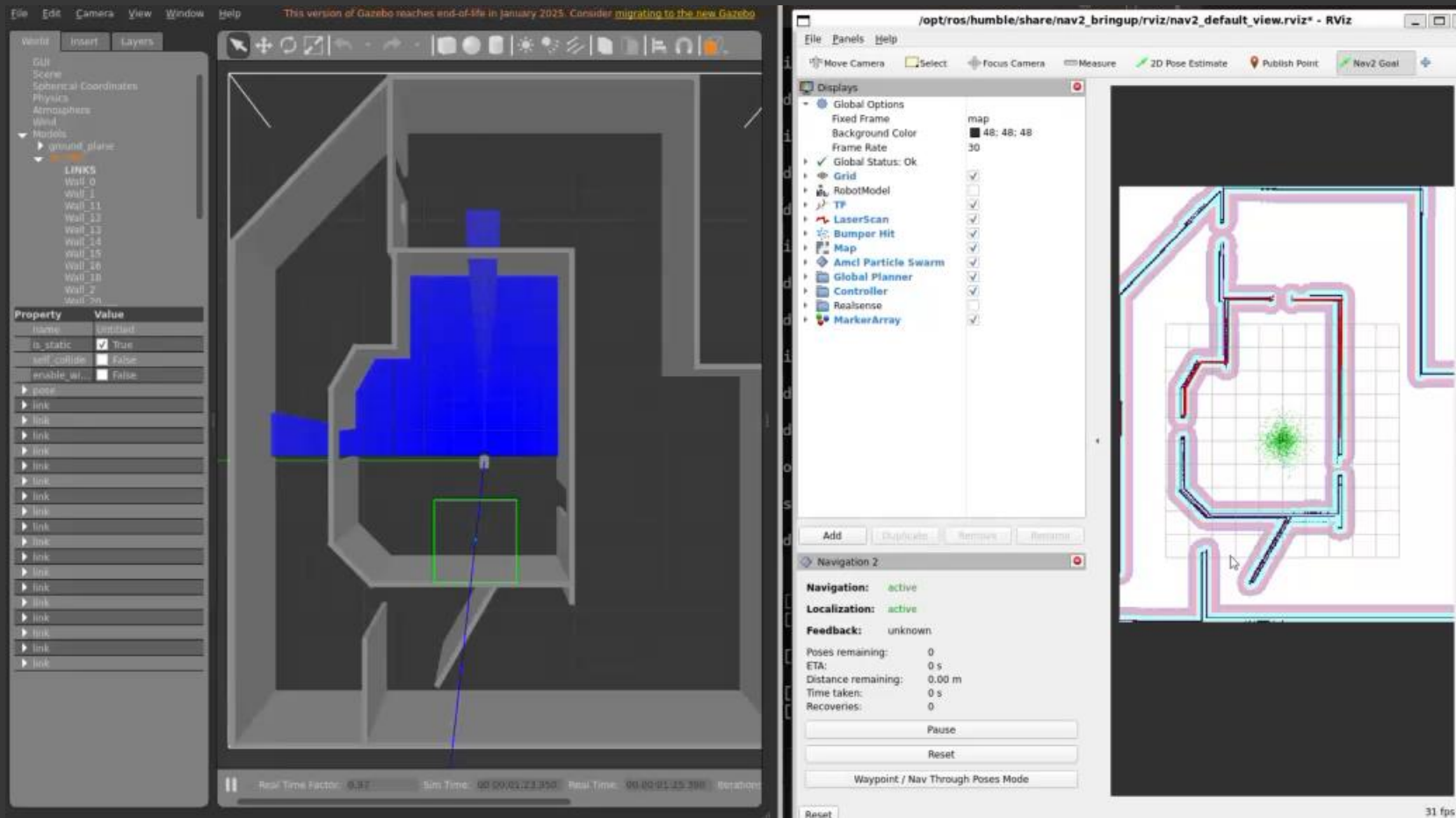
A graph-based path-planning algorithm that finds a route by expanding nodes based on cost and a heuristic estimate.

Dynamic window approach

- Uses velocity space instead of position space.
- Commands consider the robot's acceleration
- Creates smoother paths
- Lightweight algorithm



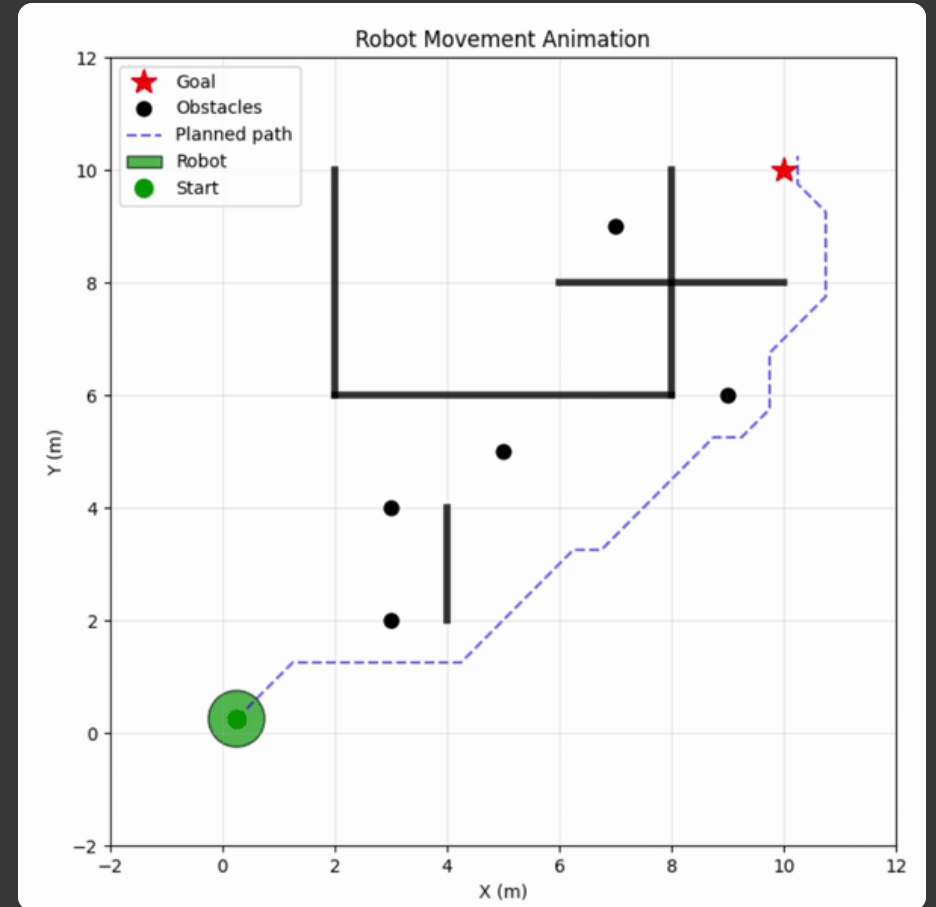
DWA – Navigation



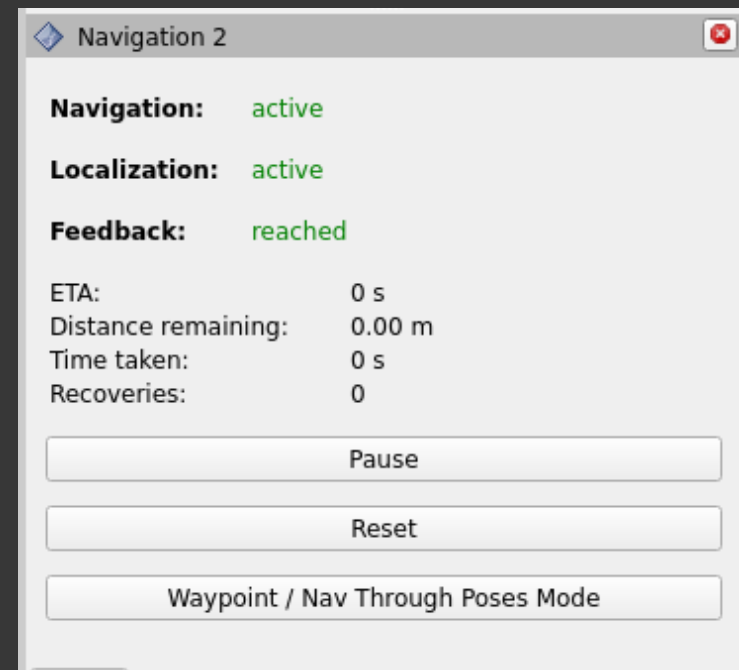
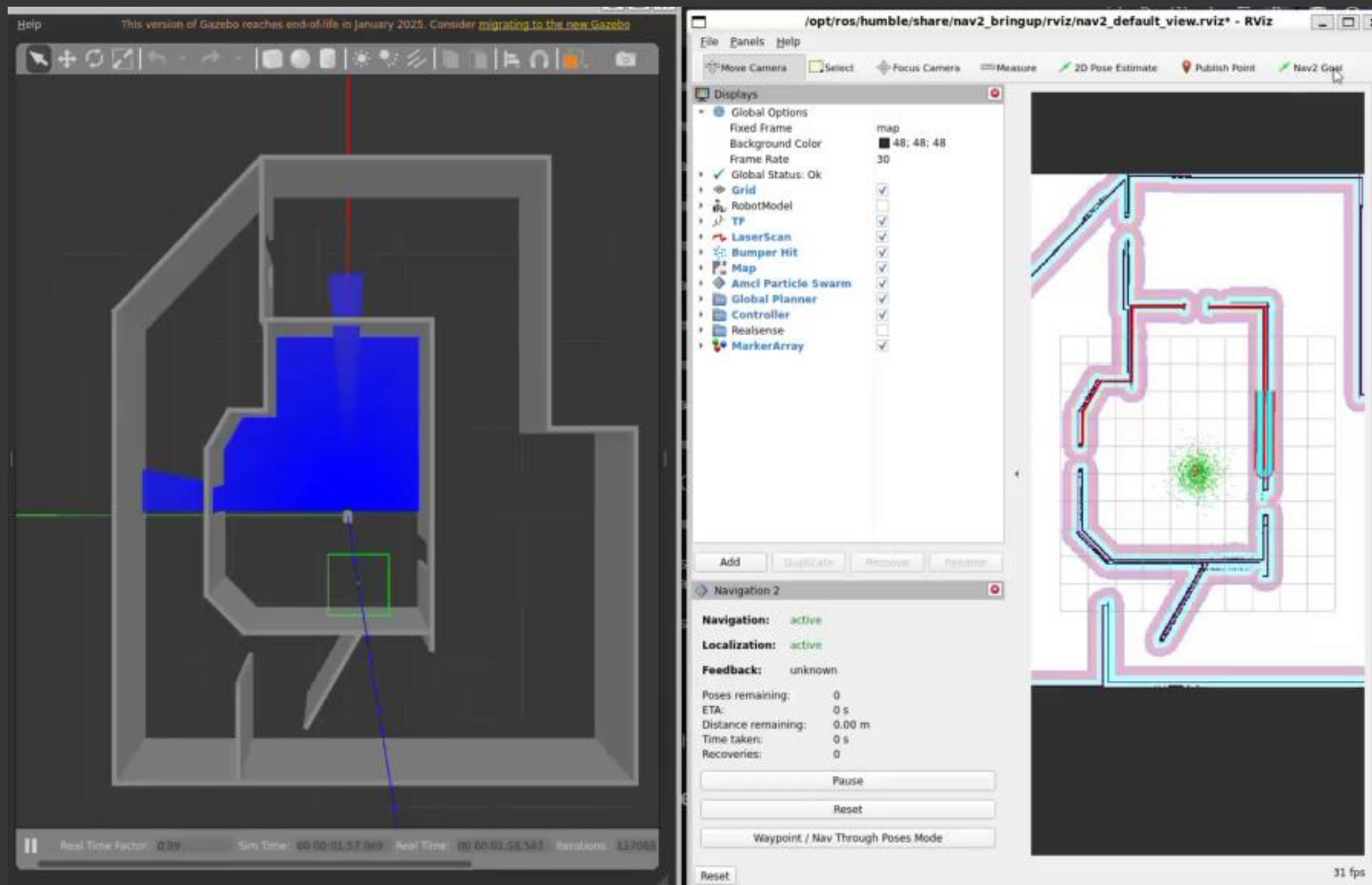
*Video
speed up*

A star path planning

- A graph-based path planning algorithm.
- Uses both the cost so far and a heuristic estimate of remaining cost.
- In the same conditions, A^* always picks the same route.
- Hugs walls, unlike DWA.



A Star - Navigation



Video speed up

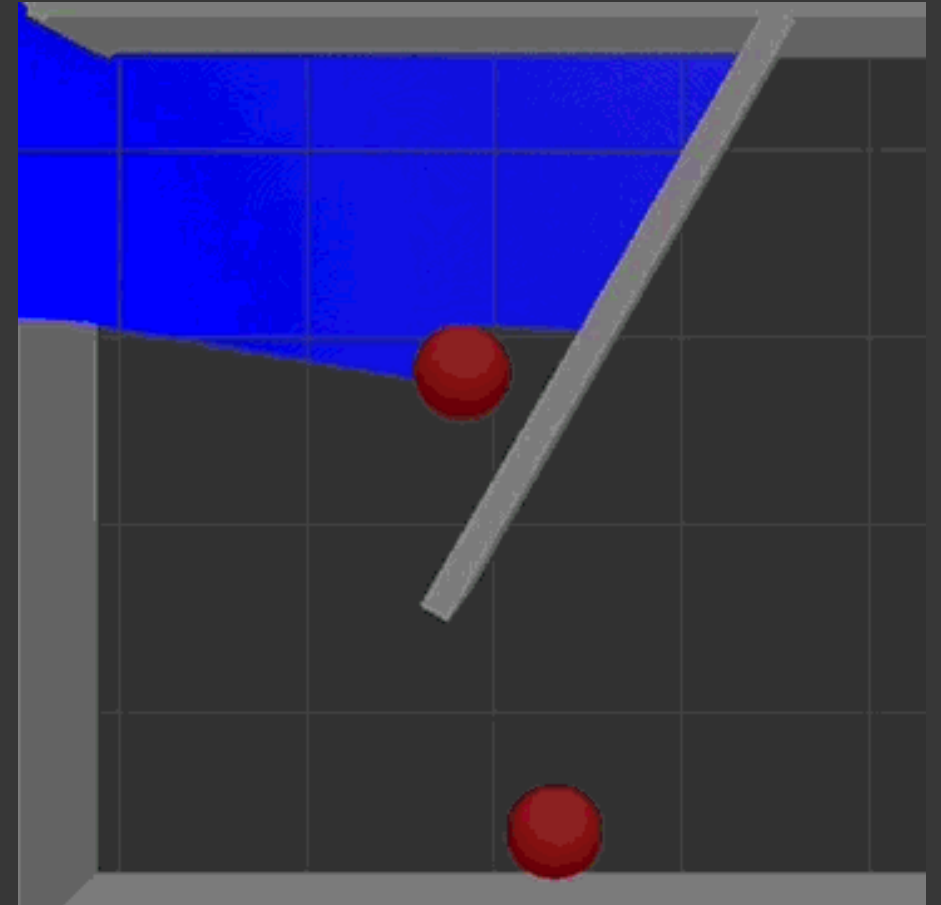
Dynamic obstacles

Within the real world, things don't always stay the same.

For example:

- People/animals walk past
- People move where things are
- Doors get closed

The LiDAR sensor detects the object, and then the nav2 stack pauses the robot if the robot is about to hit it, and, where needed, creates a new route.



Safety behaviours

- If the robot encounters a situation it does not expect
- For example, an unmapped obstacle will do one of the following:

Waiting

Stopping in place.

Back up

Reversing in a like

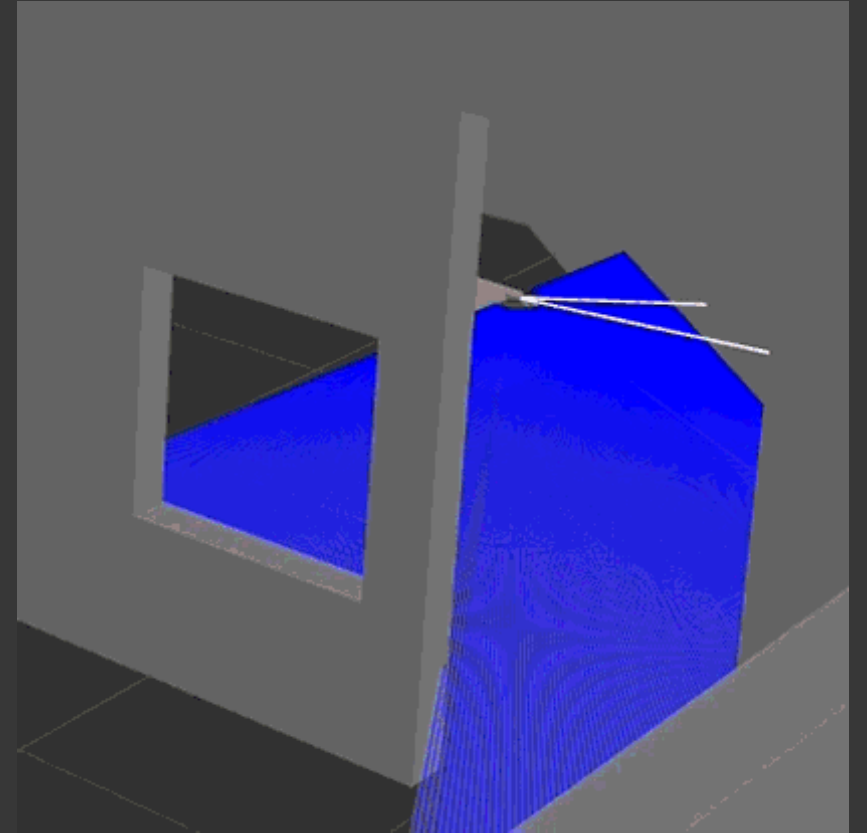
Spinning

Turning in place

Safety behavior - SPIN

- The robot spins in place.
- Allows the robot to 'look around' and see if it is trapped or if there is a viable route.
- Or can attempt to dislodge itself if stuck

Parameter	Use	Value
Spin_vel	How fast the robot can spin in radians per second	0.2
Spin_dist	How far the robot can spin in total. In radians	3.14159

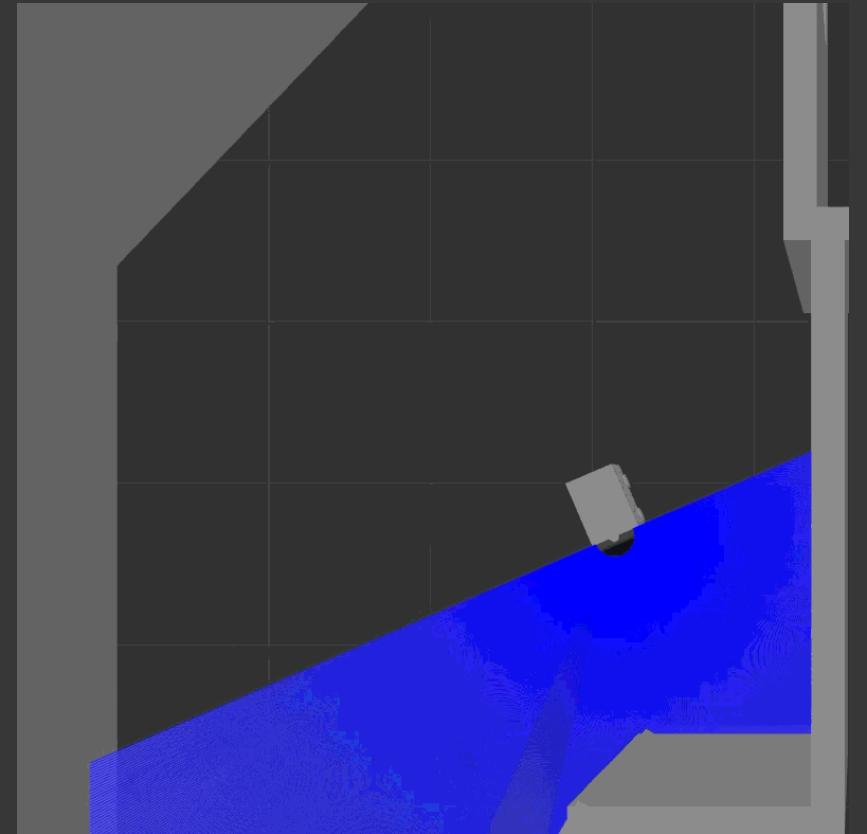


Safety behavior - Waiting

- The robot stops and waits in place.
- Allows time for the environment to change.

For example, the moving obstacle to move.

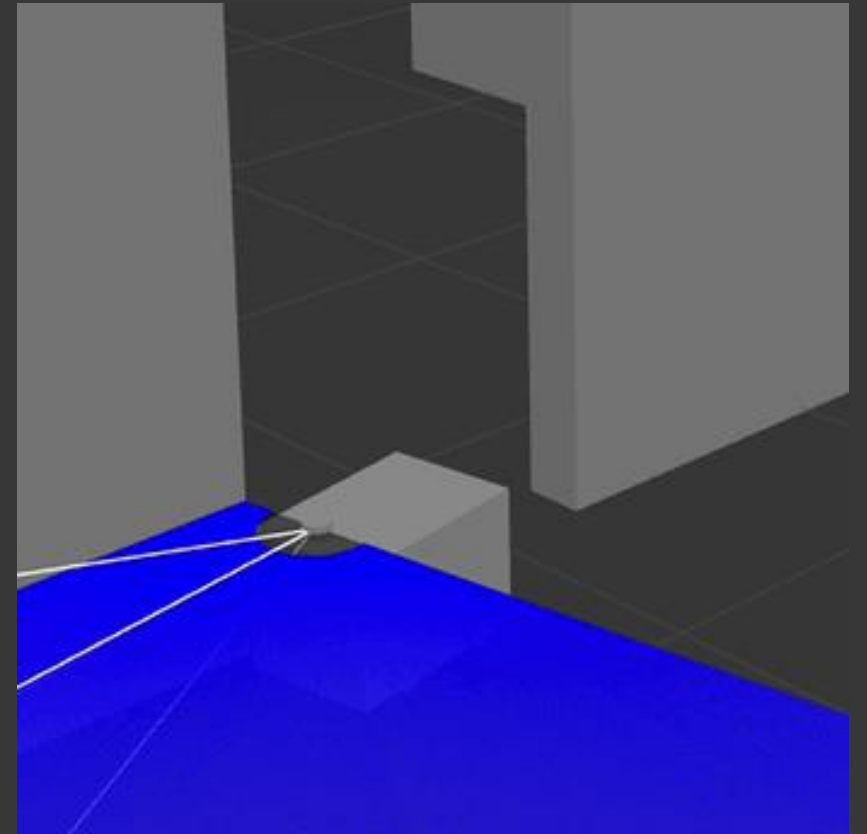
Parameter	Use	Value
Wait_ duration	The amount of time the robot will stop and wait before moving again.	2s



Safety behavior - Reversing

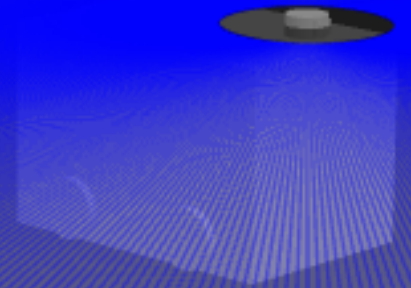
- The robot backs up.
- Allows the robot to remove itself from an enclosed space
- Backs up in a straight line

Parameter	Use	Value
Backup_vel	The speed (velocity) in which the robot can reverse in meters per second.	0.15
Backup_dist	The max distance the robot can back up in meters	0.6



Further improvements

- Investigate other path planning algorithms for both local and global navigation.
- Increase LiDAR radius to 360 degrees from its current 180.
- Increase the robots speed, while ensuring it stays safe.



THANK YOU

